QuantumTech Computing Services

Scenario

QuantumTech is a renowned computer manufacturer firm known for pushing the boundaries of bespoke computer system development. The firm is based in the heart of Techville, a fast growing and bustling hub of innovation where they have found success in servicing the computing needs of new start-up technology-based companies. The firm is well known in the area for the performance and reliability of their high-powered desktop computer systems. Recently, the firm previewed their QuantumX, a new desktop-based computer system which embodies the pinnacle of hardware and software integration, boasting huge processing power and efficiency with state-of-the-art components and a custom operating system. With new businesses continuing to flood the local area there has been an unparalleled demand for the new QuantumX, with pre-orders far exceeding QuantumTech's expectations. Additionally, the firm is also increasingly asked whether they can provide other services that go beyond computer system building. You have been hired as a business technology analyst to report on the computing services currently offered by the firm and provide insight into the future direction of QuantumTech. The firm is keen to expand into different areas and you recognise the need for them to broaden their range of computing services by offering other classes of systems, software, and networking. You are tasked with reporting on the various avenues for expansion with a focus on the underlying technologies. The report will be used to inform a new proposal for financial backing needed to expand QuantumTech's service

Task 1

(a)

What is a CPU ? .. CPU in general is considered the brain of the computer: “A Central Processing Unit is the most important component of a computer system. A CPU is hardware that performs data input/output, processing, and storage functions for a computer system. A CPU can be installed into a CPU socket. These sockets are generally located on the motherboard. CPU can perform various data processing operations. CPU can store data, instructions, programs, and intermediate results.” (geeksforgeeks.org).

The benefits of the CPU

1- The CPU has the ability to handle complex tasks and manages everything from basic calculation to running the operation system.

2- The CPUs are capable of performing billions of calculations per second, which means that they are very fast. Especially the modern CPUs

3- The CPUs have multiple cores, which allow them to handle multiple tasks at the same time.

4- users can run different applications with one CPU because the CPUs are designed to work with various types of software. (geeksforgeeks.org)

For the desktop computer QuantumX, it is very important to have a strong CPU. since the CPUs are capable of performing billions of calculations in a short period of time. That means CPUs are super fast and can especially benefit QuantumX in running complex business applications and data analysis.

Key features of the CPU such as clock speed improves the overall performance of the computer depending on the speed limit. Higher the speed, the more it can improve the performance of the computer. Additionally, cache memory is another key feature in CPUs. According to itamg.com, “It helps reduce the time taken to fetch data from the main memory, improving the overall performance of the CPU.” Which helps in reducing the time needed to access data that is used frequently.

Furthermore, Having multiple cores within the CPU helps in handling multiple tasks at the same time: “Single-core CPUs have one core, while multi-core CPUs like dual-core, quad-core, hexa-core, octa-core, and deca-core processors have multiple cores, enabling them to handle more tasks simultaneously.” (itamg.com).

While CPUs help in faster performance of the computer, there are other main components of the hardware such as Storage and RAM that also provide special features to ensure a high performance desktop computer.

Storage is one of the most important components in a computer as it is responsible for storing data: “Computer storage refers to components that provide long-term data retention, such as hard disk drives (HDDs) and solid-state drives (SSDs).” (hp.com)

(HDD) refers to Hard Disk Drive and (SSD) refers to Solid State Driver are devices that store data. But each device stores data differently.

According to aws.amazon.com, “Solid state drives (SSD) and hard disk drives (HDD) are data storage devices. SSDs store data in flash memory, while HDDs store data in magnetic disks. SSDs are a newer technology that uses silicon's physical and chemical properties to offer more storage volume, speed, and efficiency.”

Solid state drive is considered a new technology. It is faster and more durable than a Hard Disk Drive. Solid State Drive is “more durable because it does not contain optical disks inside like hard disks. It needs less power as compared to hard disks, is lightweight, and has 10x faster read and writes speed as compared to hard disks. But, these are costly as well.” (geeksforgeeks.org).

Moreover, Hard Disk Drive is “a storage device (HDD) that stores and retrieves data using magnetic storage. Most computers and laptops have HDDs as their secondary storage device. The read-write speed of HDDs is not so fast but decent. It ranges from a few GBs to a few and more TB.”
(geeksforgeeks.org)

The QuantumX is equipped with modern storage options, which means that both Solid State Drives and Hard Disk Drive are installed in QuantumX. In terms of SSD, SSD provides faster read/write speed. Which enables fast boot and load time. However, SSDs are usually more expensive than HDDs. in contrast, HDDs offer larger storage capacity and they are less expensive. In my opinion, These technologies can work together to give the QuantumX a speed and capacity balance that makes it perfect for companies that need quick access to big data volumes.
(avast.com).

Another key component in a computer's hardware is RAM. RAM refers to Random Access memory, which is “the hardware in a computing device that provides temporary storage for the operating system (OS), software programs and any other data in current use so they're quickly available to the device's processor.” (techtarget.com).

It is highly recommended to give both RAM capacity and speed top priority for the QuantumX computer. The amount of data that can be stored and accessible at once is determined by the RAM capacity, which is expressed in gigabytes (GB) or terabytes (TB). For operations like virtualization, gaming, and video editing, a larger RAM capacity is necessary to allow the system to manage several processes effectively.

Furthermore, the speed at which data can be read from or written to the memory is affected by the RAM speed, which is expressed in megahertz (MHz) or gigahertz (GHz). Increased RAM speeds improve system responsiveness by allowing data flow between the RAM and the CPU.
(library.mosse-institute.com).

In order to reach the desired growth for QuantumTech, I would suggest expanding in two areas. Which are mainframe computer and networking services.

Mainframe is a strong computer that is used by large organizations. According to ibm.com, “mainframes are high-performance computers with large amounts of memory and data processors that process billions of simple calculations and transactions in real-time. A mainframe computer is critical to commercial databases, transaction servers and applications that require high resiliency, security and agility.”

Why are mainframe computers recommended for QuantumTech??
Since mainframe computers contain large amounts of memory and data processors that are capable of processing billions of calculations with a high speed, that means this kind of computers could be attractive for companies seeking high computing solutions. However, mainframe computers are often very expensive and could not be suitable for small businesses as it is used for large companies.
(hasonss.com)

Networking services on the other hand “refer to the applications or services that are hosted on a network to provide functionality for users or other applications.” (nilesecure.com).

Why are networking services important??

According to the same page, Network services are essential for technologically dependent modern enterprises. They enhance resource utilization and increase output. Fast and dependable networks are essential for efficient operations and company expansion. Significant delays and financial losses can result from outages and poor connections.

Constant connectivity is guaranteed by a robust network architecture, which is essential for both short-term and long-term objectives. Beyond simply connecting devices, network services enable businesses to fully leverage technology for improved outcomes in a variety of sectors.

This means that computers with high networking services are demanded for business networks, which increases the revenue of QuantumTechs.
For businesses however ,setting it up could be expensive because the cost of the wires and cables are high and require skilled people to manage the network. (geeksforgeeks.org)

(b)

For QuantumTech, it is very important to have a Modern Operating system within its computers for managing resources and to ensure computers to run smoothly. Modern Operating Systems have 3 key important functions such as Device management, user interface, and networking.

1- Device management: Drivers are used by the OS to control device connectivity. It manages requests from linked devices, tracks them, and distributes access effectively.

2- User interface: The OS offers a graphical user interface (GUI), batch interface, or command-line interface (CLI) as its user interface (UI). These user interfaces make it simple for users to interact with the system.

3- Networking: By enabling networking, the operating system makes it possible for the computer to connect to other devices and systems. It provides interfaces, security features, and network protocols.

(geeksforgeeks.org)

Task 2

(a)

For QuantumTech, there are three types of software that need to be considered for QuantumTech's desktop computer (QuantumX) such as Application Software, System Software, and Development Software.

Application software (app) is the software that allows the user to perform a specific task. According to geeksforgeeks.org, "application software is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements. It includes word processors, spreadsheets, database management, inventory, payroll programs, etc."

For example, word processing and spreadsheets applications include Microsoft Office (MS Word, MS Excel, Powerpoint). Which can help businesses in performing daily work tasks.

Additionally, system software is "software that directly operates the computer hardware and provides the basic functionality to the users as well as to the other software to operate smoothly. Or in other words, system software basically controls a computer's internal functioning and also controls hardware devices such as monitors, printers, and storage devices, etc." (geeksforgeeks.org).

For example, Operating systems include Windows, Linux, and macOS.

While communication and connection software include Device driver.

Operating systems like Windows, Linux , and macOS ensure fast, smooth operations and manage resources.

Furthermore, Development software or (programming software) are “programs that are used to write, develop, test, and debug other software programs. It’s helpful to think of these programs as a translator of sorts: they take programming languages like Laravel, Python, C++, and more and translate them into something a computer or phone will understand.” (leadwithprimitive.com).

Programming software helps developers create applications using programming languages such as Python,C++, and java.

Usually, new computers come with free installed applications for enhancing user experience. For QuantumX, it is important to adopt this action for a more competitive place in the market. There are four types of application software such as productivity, multimedia, communication, and entertaining software.

According to en.wikipedia.org, “Productivity software (also called personal productivity software or office productivity software[1]) is application software used for producing information (such as documents, presentations, worksheets, databases, charts, graphs, digital paintings, electronic music and digital video).”

For example, businesses rely on most productivity softwares like Microsoft Office, which includes (Word, Excel, PowerPoint) and Google Docs for writing, editing, and data analysis.

In addition, Multimedia software is a “software that plays or records and creates audio and video files.” (computerhope.com).

According to tutorchase.com, “Multimedia software applications are designed to handle and manipulate multimedia data, which includes text, sound, graphics, video, and animation. These applications are essential tools in the digital world, used in a wide range of fields such as education, entertainment, business, and science. They enable users to create engaging and interactive content, enhancing communication and understanding.”

Multimedia softwares are used for video production purposes and entertainment such as video games. (computerhope.com)

Communication Software on the other hand is “used to provide remote access to systems and exchange files and messages in text, audio and/or video formats between different computers or users. This includes terminal emulators, file transfer programs, chat and instant messaging programs, as well as similar functionality integrated within MUDs.” (en.wikipedia.org).

For example, there are various communication softwares such as Zoom, Google Meet, Microsoft Teams,whatsapp, and many other famous communication softwares. Communication softwares became essential in our modern world. Communication softwares has transformed both personal and professional life by enabling remote work and online education.

Entertainment software is “a type of software that typically has more relaxed security requirements compared to sensitive applications. It focuses on delivering video content with high quality and timely updates, often offering high-quality content for a price while low-quality versions may be free to the public.”(sciencedirect.com).

Examples of entertainment softwares are Netflix for watching movies, Steam for installing video games, and Spotify for listening to music.

QuantumTech are not familiar with how software works with computer hardware. Hence, I will explain how these components work together.

According to math.hws.edu (Javanotes 9, Section 1.1), “The CPU executes a program that is stored as a sequence of machine language instructions in main memory. It does this by repeatedly reading, or fetching, an instruction from memory and then carrying out, or executing, that instruction. This process is called the fetch-and-execute cycle. ”

(b)

What is a compiler ?

A compiler is “software that translates or converts a program written in a high-level language (Source Language) into a low-level language (Machine Language or Assembly Language).” (geeksforgeeks.org)

There are many types of compilers, including GNU Compiler and Microsoft C ++ .

Compilers allow programmers to provide particular options about the hardware that the program may operate on, which can enhance program performance. This makes it possible for the compiler to produce machine code that increases the efficiency of the chosen hardware. (ca.indeed.com, june 24, 2024)

What is an interpreter ?

“An interpreter is a program that directly executes the instructions in a high-level language, without converting it into machine code.”

Examples of interpreters include python interpreter and javascript interpreter. (lenovo.com)

Because an interpreter reads the code line by line, it makes debugging the code easier. Additionally, it facilitates debugging and code modification for programmers. (ca.indeed.com, june 24, 2024)

What is IDE ?

“An integrated development environment (IDE) is a software application that helps programmers develop software code efficiently. It increases developer productivity by combining capabilities such as software editing, building, testing, and packaging in an easy-to-use application.”
(aws.amazon.com).

Examples of an Integrated Development Environment include pycharm, and microsoft visual studio. (indeed.com, july 8, 2023)

Programmers can write code more quickly and effectively by utilizing an IDE, which is one of its primary advantages. An IDE offers a number of tools and features to help with coding, including syntax highlighting, code editing, code completion, formatting, and refactoring. (linkedin.com).

What is Vision Control ?

“Version control systems are a category of software tools that helps in recording changes made to files by keeping a track of modifications done in the code.” (geeksforgeeks.org)

Most famous examples of Vision Control include Git, Helix core, and Microsoft TFS. (geeksforgeeks.org)

The Version Control System records all code changes, which aids the development team in managing the source code. Additionally, it protects against any unexpected human error and its effects on the source code.

(simplilearn.com).

Task 3

(a)

What is LAN ??

LAN refers to Local Area Network. Which is “a collection of devices connected together in one physical location, such as a building, office, or home.” (cisco.com)

How does it work ?

It works by connecting devices within the network using Ethernet, WiFi, or both. (cloudflare.com)

The advantages of LAN:

1- it is easy to set up and cost effective (geeksforgeeks.org).

2- Users can view and even control each other's devices, share files, print to shared printers, and connect to a single Internet connection.
(cisco.com).

The disadvantages of LAN:

The main disadvantage of a LAN is that it often only covers a single building or campus. (testbook.com).

What is WAN ?

WAN refers to Wide Area Network: “a wide-area network (WAN) is a collection of local-area networks (LANs) or other networks that communicate with one another.” (cisco.com).

How does it work ?

WANs allow devices from all over the world to communicate, share information and do much more by the help of WAN provider. (comptia.org).

The advantages of WAN

WANs help in improving communication between separated offices, enhancing cooperation, and centralizing data management.
(shiksha.com)

The disadvantages of WAN

1- Maintaining the network is challenging and requires knowledgeable technicians and network administrators.

2- WAN troubleshooting can be difficult and time consuming. (shiksha.com).

What is MAN ?

MAN refers to Metropolitan Area Network: “a computer network that connects computers within a metropolitan area, which could be a single large city, multiple cities and towns, or any given large area with multiple buildings. A MAN is larger than a local area network (LAN) but smaller than a wide area network (WAN).” (cloudflare.com).

How does MAN work ?

It works by connecting LANs together through fiber optic cables. (cloudflare.com)

The advantages of MAN:

1- allows several places in a city to be connected seamlessly

2- enables faster communication and data transfer

3- reduces costs by combining network resources. (lighyear.ai)

The disadvantages of MAN:

1- A metropolitan area network requires more cable to set up than a local area network (LAN)

2- Network administrators and technicians with the necessary skills must set up these networks

3- The costs of setting up and maintaining this network are more than those in a local area network. (shiksha.com).

For QuantumX computers, and since QuantumTech are planning to expand their services, I would recommend using and implementing both LAN (Local Area Network) and MAN (Metropolitan Area Network) because MAN allows QuantumTech to connect a large number of offices within a building, city, or country. Additionally, the existence of LAN in each office allows employees to collaborate with each other easily through remote access and file sharing without the need of USB. Since LANs are easy to configure, That means QuantumTech could offer setup services as part of their expansion plan.

(b)

What is Star topology ??

Star topology is a structure where all the devices are connected to a central device, such as the router , hub, and switch . It allows the data to flow through the central device, helping in managing the communication between the nodes. (geeksforgeeks.org) (techtarget.com)

The advantages of star topology:

1- it is very reliable even if one cable goes down

2- it ensures high performance

3- Affordable for small networks.

The disadvantages of star topology:

1- it requires more cables, unlike the bus topology.

2- if the central device gets damaged, the devices will no longer work.

3- constant maintenance is required for the central device to work properly. (geeksforgeeks.org).

What is tree topology ?

Tree topology is “a hierarchical network topology resembling a tree's branches. It combines characteristics of both star and bus topologies.” (shiksha.com).

The advantages of tree topology:

1- This topology is a hybrid of star and bus topologies

2- If one node in a network is damaged or stops functioning, the other nodes in the network are unaffected

3- This topology offers significant scalability since the leaf nodes can add one or more nodes to the hierarchical chain (geeksforgeeks.org, 28 dec, 2024).

The disadvantages of tree topology:

1- When compared to other network topologies, this network can be very hard to configure.

2- it needs more cables than the star and ring topologies. (geeksforgeeks.org).

What is mesh topology ?

Mesh topology is a structure where all the devices are interconnected, providing multiple paths for data to travel. (geeksforgeeks.org).

The advantages of mesh topology:

1- very reliable and it provides high security and privacy

2- new devices can be added easily without major changes

3- if the connection of a device fails, data can still be transmitted through alternative paths.

The disadvantages of mesh topology:

1- it is expensive because it requires more cables and network devices.

2- it is difficult to set up due to a large number of devices that need to be interconnected.

3- it is difficult to troubleshoot due to the several paths that can be used by data. (geeksforgeeks.org 25 oct, 2024).

What is Hybrid topology ?

Hybrid topology is “defined as a network topology that combines two or more different network topologies. A hybrid topology can be a combination of bus topology, ring topology and mesh topology.” (geeksforgeeks.org , 25 oct, 2023).

the advantages of hybrid topology

1- In hybrid topologies, adding a new node or removing an existing node is simple

2- Compared to individual star, ring, and mesh topologies, hybrid topologies are more scalable, secure, and reliable

3- Hybrid topology provides high performance and speed. (geeksforgeeks.org, 25 oct, 2023).

the disadvantages of hybrid topology

1- It is challenging to design and implement a hybrid network topology

2- For a hybrid topology, more wires and other hardware components are needed

3- Installing a hybrid topology is a challenging process (geeksforgeeks.org, 25 oct, 2023).

Task 4

A)

What does the term (distributed computing) mean ?

According to aws.amazon.com, “Distributed computing is the method of making multiple computers work together to solve a common problem. It makes a computer network appear as a powerful single computer that provides large-scale resources to deal with complex challenges.”

How does distributed computing work ?

According to the same website, “Distributed computing works by computers passing messages to each other within the distributed systems architecture. Communication protocols or rules create a dependency between the components of the distributed system. This interdependence is called coupling.”

A lot of companies use distributed computing, including cloud computing platforms such as Amazon Web Services, Google Cloud, and Microsoft Azure. There are two types of distributed computing applications. Which are Client server model and peer to peer model

Client server model is “a distributed application structure that partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters called clients.”

How does it work ?

It works by one server serving several clients under the client server model. Consequently , the server handles many requests from various clients. (ionos.com, Jan 31, 2023).

The advantages of Client server model:

1- It provides a centralized control system.

The disadvantages of Client server model:

1- Clients are vulnerable to worms, Trojan horses, and viruses if they are uploaded to or present on the server.

Peer to peer model is “a decentralized network architecture in which participants, called peers, interact directly with each other without needing a central authority or server. In a P2P network, each participant acts as both a client and a server, enabling them to share resources and services directly with other peers.” (spiceworks.com, november 7, 2023).

How does it work ?

According to techtarget.com, “each computer has the same responsibilities and capabilities. Since there is no server, the computers connect with each other in a workgroup to share files, printers and internet access. This architecture is practical for workgroups of 12 or fewer computers.”

the advantages of peer to peer model

1- it is easy to set it up

2- cost effective

The disadvantages of peer to peer model

1- it is less secure because the computer can be accessed anytime

2- it requires backups on each device individually. (infosecawarenss.in)

For QuantumTech, I would recommend using client server model because client server model allows centralized control, which helps QuantumTech servers take full control of all computing tasks. Making it easier to monitor, update, and secure their servers.

B)

What is Quantum computing ?

According to ibm.com, “Quantum computing is an emergent field of cutting-edge computer science harnessing the unique qualities of quantum mechanics to solve problems beyond the ability of even the most powerful classical computers.”

How does quantum computing work ?

“Quantum computers perform calculations based on the probability of an object's state before it is measured - instead of just 1s or 0s - which means they have the potential to process exponentially more data compared to classical computers.” (sciencealert.com).

the advantages of Quantum computing:

1- Compared to traditional computers, quantum computers can execute calculations at an enormously quicker rate.

2- Quantum memory devices provide effective data storage and retrieval in quantum computing.

3- Google searches and other information retrieval methods could be completely transformed by quantum computing. (theknowledgeacademy.com, sienna roberts, 21 february 2025).